

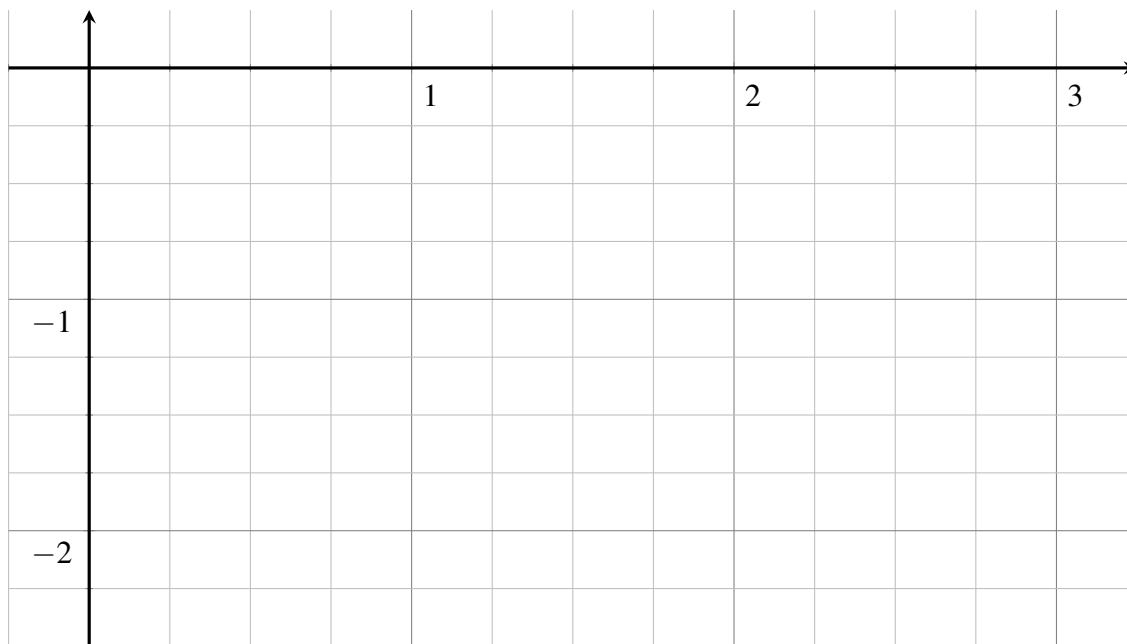
Learning target DF1, version 5

Estimate the value of a derivative using difference quotients, and draw corresponding secant and tangent lines on the graph of a function.

Suppose that you know the following values of some function $f(x)$:

x	1	1.5	2
$f(x)$	-0.8	-1	-1.6

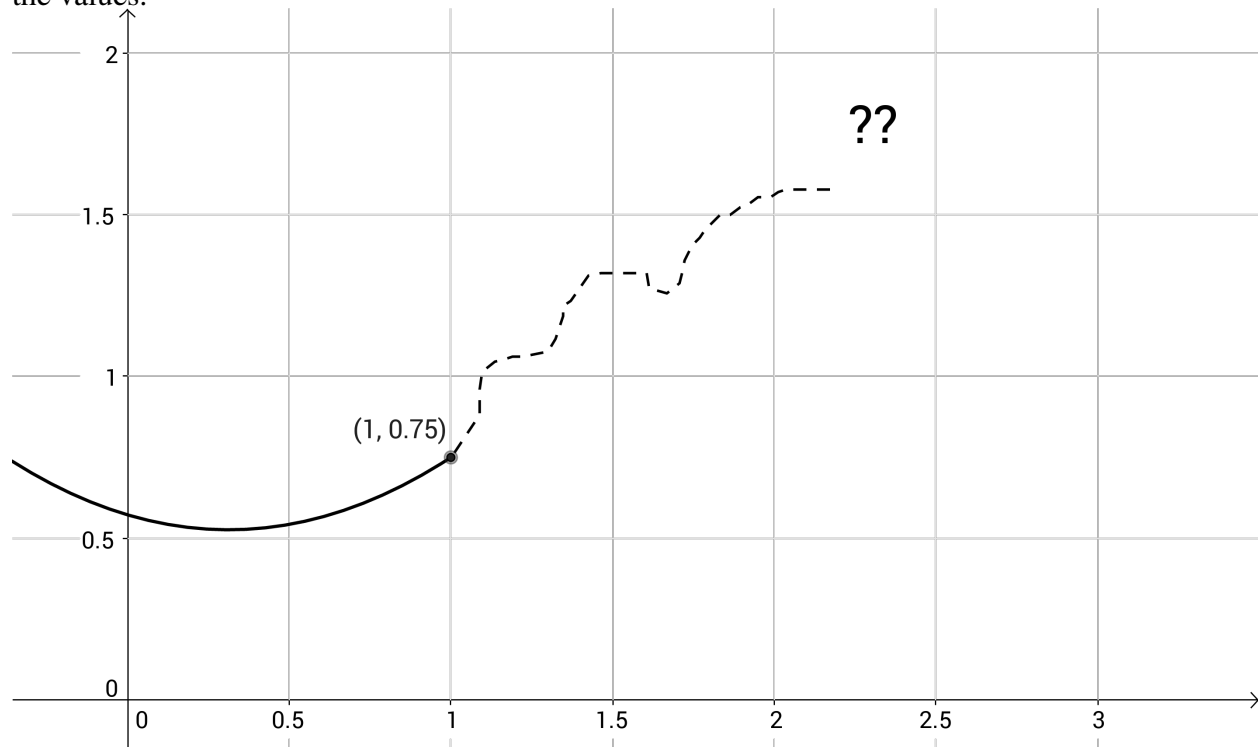
1. Plot these points and sketch a plausible graph of $f(x)$.



2. On your graph above, draw a plausible tangent line to the graph of $f(x)$ at $x = 1.5$.
3. Use a central difference to estimate $f'(1.5)$. Draw the corresponding secant line on your graph.
4. Compute another, different estimate of $f'(1.5)$. Draw the corresponding secant line on your graph.
5. Which one of your approximations is the best? How do you know?

Learning target AD2, version 4

Part of the graph of this function $f(x)$ has been erased. We'll use linear approximation to recover the values.



- (a) Draw the tangent line to this function at $x = 1$.
- (b) Suppose we know that $f'(1) = 0.65$. Use point-slope form to write down an equation for $L(x)$, the tangent line to this function at $x = 1$.
- (c) Use your equation for $L(x)$ to estimate $f(1.4)$.
- (d) Do you think it is more likely that this is an overestimate or an underestimate? How come?

Learning target DF5, version 4

Compute derivatives using the Chain Rule.

Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (chain, product, quotient, sum and difference, etc.) you are using.

1. $g(t) = 3 \cos(-3t^2 - 3t + 4)$

2. $h(y) = 5 \sin(y^{7/2})$

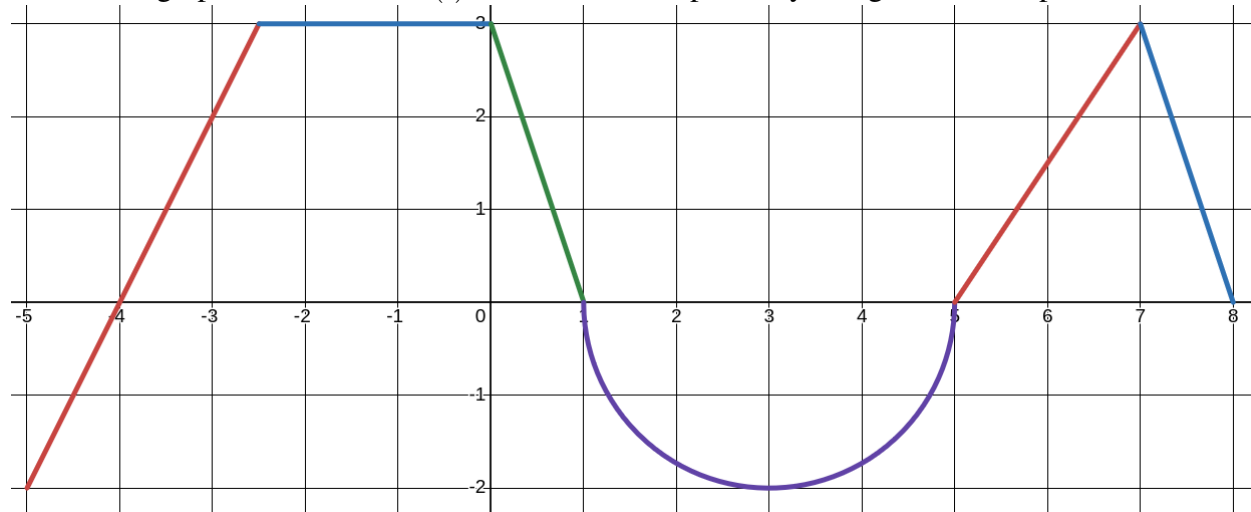
3. $k(w) = 5 \left(\sin(w) \right)^{7/2}$

4. $f(x) = -(5x + 3e^x - 4)^5$

Learning target IN1, version 4

Use geometric formulas to find definite integrals.

Here is the graph of a function $N(t)$, which is made up of only straight lines and parts of circles.



Compute each of the following.

(a) $\int_{-5}^{-4} N(t) \, dt$

(e) $\int_1^5 N(t) \, dt$

(b) $\int_{-4}^{-2.5} N(t) \, dt$

(f) $\int_5^7 N(t) \, dt$

(c) $\int_{-2.5}^0 N(t) \, dt$

(g) $\int_7^8 N(t) \, dt$

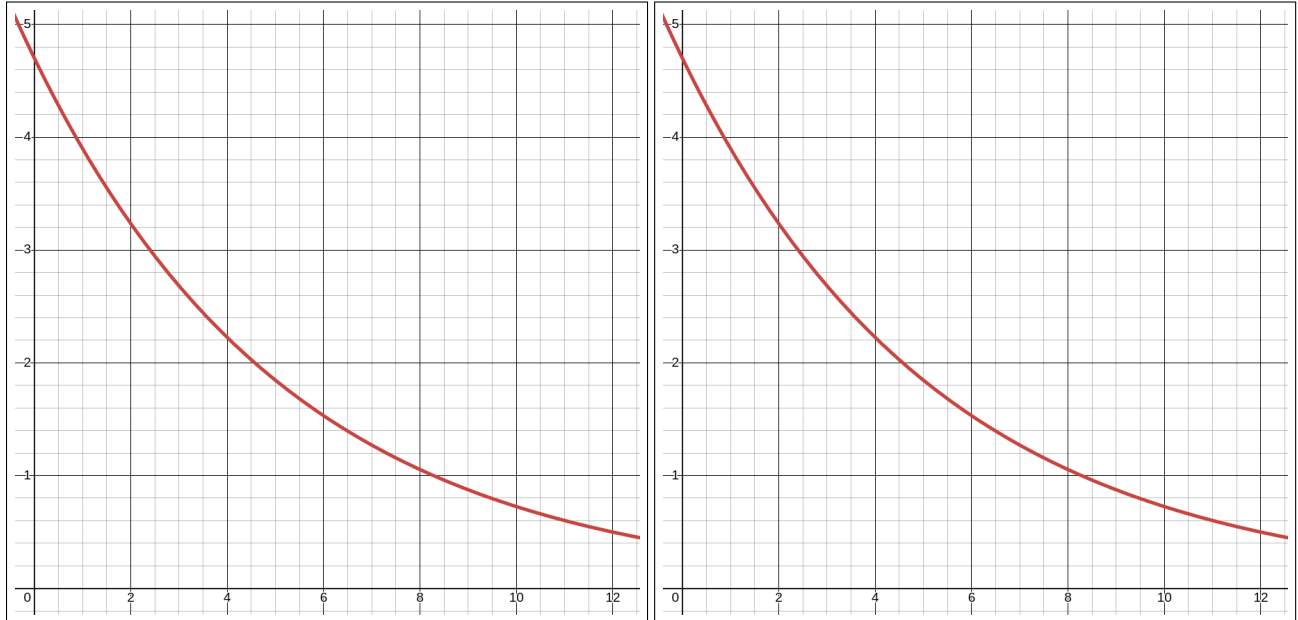
(d) $\int_0^1 N(t) \, dt$

(h) $\int_{-5}^8 N(t) \, dt$

Learning target IN2 and INx, version 3

Over the first year of its life, a golden retriever puppy gains weight at a rate of

$$p(t) = 4.7e^{-0.187t} \text{ kilograms per month.}$$



- On the first graph above, carefully draw the *left* Riemann sum with four rectangles of uniform width. (How wide should each rectangle be?)
- On the second graph above, carefully draw the *right* Riemann sum with four rectangles.
- Compute both of these Riemann sums. Show your work for computing the heights of the rectangles. (Hint: lots of your work will overlap!)

Include units on every number you write.

- Give your best guess for $\int_{t=0}^{12} p(t) dt$, **include units**, and explain what this number means about the puppy.